Aaron Howard

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Education

University of Illinois Urbana-Champaign — Champaign, IL
Electrical Engineering; Culmulative GPA: 3.46/4.00 — *Class of 2028*

Experience

SIG Robotics — Electrical Lead, Robotic Hand Project — January 2026 – Present
University of Illinois Urbana-Champaign — *Champaign, IL*

- Lead the electrical design for the end effector of a full-scale humanoid robot platform, owning the control PCB from schematic capture through layout and fabrication release.
- Direct the work of the hand project's electrical members, setting the KiCad and Git workflow, reviewing schematic changes, and maintaining the shared component library.
- Coordinate electrical requirements with the mechanical and controls subteams to align actuator selection, power budget, and wiring routing with the mechanical package.

Illini Formula Electric — Data Acquisition Board Subteam — August 2026 – Present
University of Illinois Urbana-Champaign — *Champaign, IL*

- Contribute to the data acquisition system for a Formula SAE Electric vehicle, working on sensor instrumentation and signal conditioning for onboard vehicle telemetry.

Projects

Five-Finger Robotic Hand Controller | *KiCad, ESP32-S3, Power Electronics* — January 2026 – Present

- Designing a custom control PCB in KiCad 10 around an ESP32-S3-WROOM-1 module to drive ten DYNAMIXEL XL330 smart servos over a shared half-duplex bus.
- Completed footprint assignment and design-rule setup for JLCPCB assembly, selecting components against real-time stock availability to avoid respins.
- Implementing a shared 3.3V TTL half-duplex communication interface, native USB-C programming, boot/reset circuitry, debugging access, ESD protection, and expansion interfaces.

Ultrasonic Direction Finder | *Analog Circuit Design* — Finished

- Built a fully analog 40 kHz ultrasonic direction-finding system with no microcontroller, from transmit oscillator through to directional output display.
- Designed a Schmitt-trigger relaxation oscillator and push-pull output buffer to drive the ultrasonic transmitter at sufficient current for usable range.
- Amplified and rectified returning signals from multiple receiver channels, then used an AD531J analog divider to compute the amplitude ratio between channels as a direction estimate.
- Implemented comparator-based threshold classification to map the ratio into discrete angular bins, indicated on a four-LED direction display.

Resistor-Key Ignition Security System | *Analog Comparators, Sensor Interfacing* — Finished

- Designed an ignition-interlock circuit that authenticates a physical key by its resistance value, rejecting keys outside a narrow tolerance band.
- Built a Wheatstone bridge front end to convert key resistance into a differential voltage with good rejection of supply variation.
- Implemented a dual-comparator window detector using LM311P devices to enable the ignition output only when the measured value falls between the upper and lower thresholds.
- Validated threshold accuracy and noise margin on the bench across the full range of decoy key values.

Technical Skills

Languages: C, C++, Python, SystemVerilog, LC-3 Assembly
Embedded & Hardware: PCB design (schematic capture and layout), analog signal conditioning, digital logic, ESP32-S3, Arduino, NVIDIA Jetson
Tools: KiCad, Vivado, Git/GitHub, SolidWorks, VS Code
Lab: Oscilloscope, function generator, digital multimeter, bench power supply, soldering and rework